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23 mins 23 questions

Multiple Choice Questions

Limits

Definition of a Limit / Infinite Limits & Limits at Infinity / Properties of Limits / Evaluating Limits Analytically / Evaluating Limits Numerically & Graphically / Squeeze Theorem & Trigonometric Limits

Easy (7 questions)	/7
Medium (9 questions)	/9
Hard (7 questions)	/7
Total Points	**/23**

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Easy Questions

1 The function f is continuous at all points except $x = 0$. The table below gives the values of f for a number of different values of x.

x	$f(x)$
-0.1	0.99833417
-0.01	0.99998333
-0.001	0.99999983
0	not defined
0.001	0.99999983
0.01	0.99998333
0.1	0.99833417

What does the table suggest that the value of $\lim_{x \to 0} f(x)$ is?

A. undefined

B. 0

C. 1

D. ∞

(1 point)

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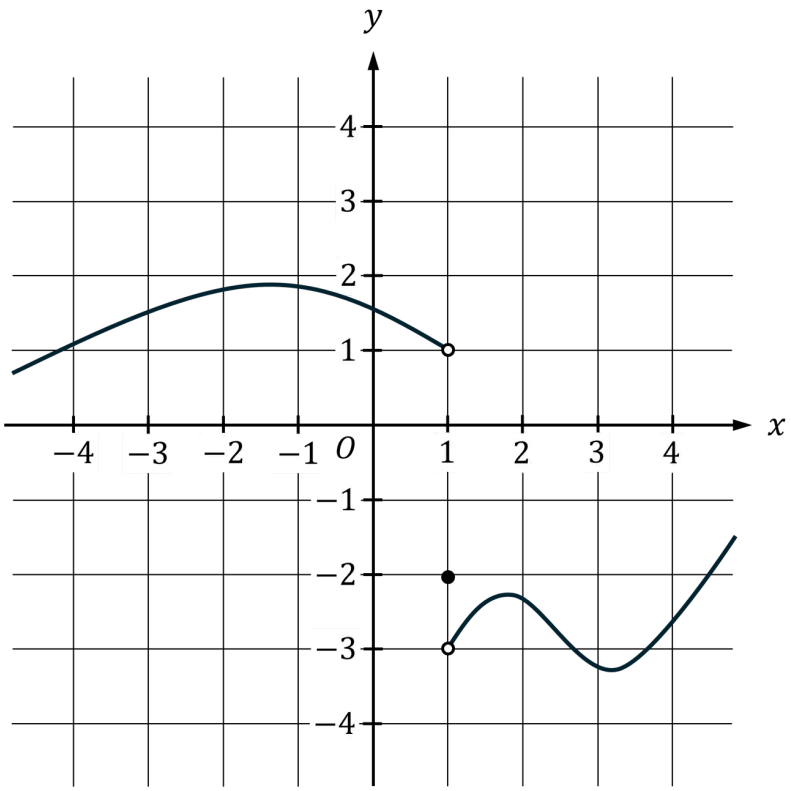
2



Graph of f

The graph of the function f is shown in the figure above. The value of $\lim_{x \to 1^+} f(x)$ is

A. -3

B. -2

C. 1

D. nonexistent

(1 point)

3 Let f and g be functions such that $\lim_{x \to 2} f(x) = 4$ and $\lim_{x \to 2} g(x) = -3$.

If the function h is defined by $h(x) = 3f(x) + g(x)$, then $\lim_{x \to 2} h(x)$ is

A. undefined

B. -5

C. 1

D. 9

(1 point)

4 The line $x = -2$ is a vertical asymptote to the graph of which of the following functions?

A. $y = \dfrac{1}{x+2}$

B. $y = x + 2$

C. $y = -2x$

D. $y = e^{x+2}$

(1 point)

5 $\lim\limits_{x \to \infty} \left(4 - \dfrac{3}{x}\right)$ is

A. nonexistent

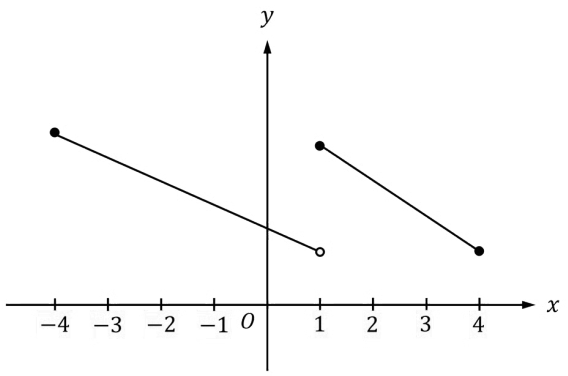
B. 0

C. 1

D. 4

(1 point)

6



Graph of f

The figure above shows the graph of a function f with domain $-4 \le x \le 4$ Which of the following statements are true?

I. $\lim\limits_{x \to 1^-} f(x)$ exists

II. $\lim_{x \to 1^+} f(x)$ exists

III. $\lim_{x \to 1} f(x)$ exists

A. I only

B. II only

C. I and II only

D. I, II and III

(1 point)

7 $\lim_{x \to \infty} \frac{2x^5 + 4x^2 + x - 3}{5x^5 + x^3 - 2x^2 + 4} =$

A. $-\frac{3}{4}$

B. 0

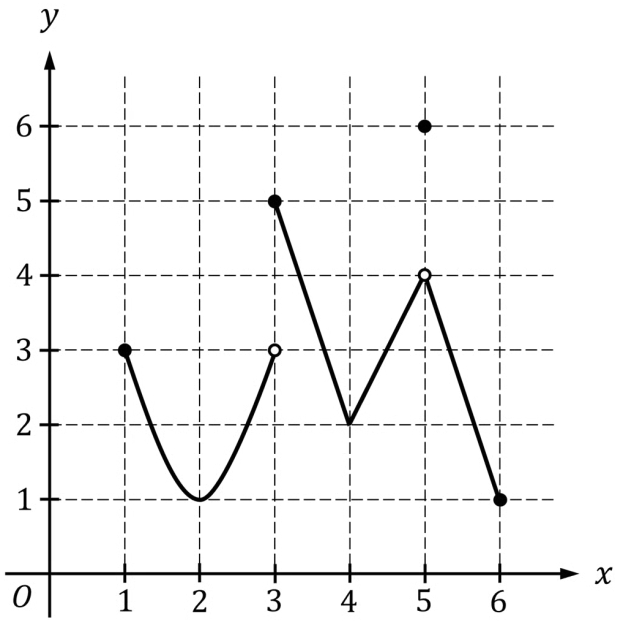
C. $\frac{2}{5}$

D. nonexistent

(1 point)

Medium Questions

1



Graph of f

The graph of the function f is shown above. Which of the following statements is false?

A. $\lim_{x \to 2} f(x)$ exists.

B. $\lim_{x \to 3} f(x)$ exists.

C. $\lim_{x \to 5} f(x)$ exists.

D. The function is continuous at $x = 4$.

(1 point)

2 The line $y = 3$ is a horizontal asymptote to the graph of which of the following functions?

A. $y = \dfrac{1}{x-3}$

B. $y = 3x$

C. $y = \dfrac{x - 3x^2}{1 - x^2}$

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D. $y = \dfrac{3x}{7-x}$

(1 point)

3 If $f(x) = \begin{cases} \ln x & \text{for } 0 < x \le 3 \\ x\ln 3 & \text{for } 3 < x \le 5 \end{cases}$, then $\lim\limits_{x \to 3} f(x)$ is

A. $\ln 3$

B. $\ln 27$

C. $\dfrac{1}{3}$

D. nonexistent

(1 point)

4 Let f be a function such that $\lim\limits_{x \to 0} f(x) = -1$.

Let g be the function defined by $g(x) = \sin x$.

What is $\lim\limits_{x \to 0^-} \dfrac{f(x)}{g(x)}$?

A. $-\infty$

B. -1

C. 0

D. $+\infty$

(1 point)

5 Let f, g and h be functions defined on the interval $1 < x < 3$. $\lim\limits_{x \to 2} g(x) = \lim\limits_{x \to 2} h(x) = -5$, and it is also true that $g(x) \le f(x) \le h(x)$ for all x such that $1 < x < 3$.

Which one of the following results allows you to determine that $\lim\limits_{x \to 2} f(x) = -5$?

A. fundamental theorem of calculus

B. intermediate value theorem

C. mean value theorem

D. squeeze theorem

(1 point)

6 $\lim_{x \to \infty} \frac{(3x-2)(1-x)}{(x-2)(x+1)}$ is

A. -3

B. 1

C. 3

D. nonexistent

(1 point)

7 $\lim_{x \to 0} \frac{7x^5 - 3x^4}{9x^5 + 2x^4}$ is

A. $-\frac{3}{2}$

B. 0

C. $\frac{7}{9}$

D. nonexistent

(1 point)

8 The domain of a function f is $x \leq 0$. The horizontal line $y = -1$ is an asymptote for the graph of f. Which of the following statements must be true?

A. $\lim_{x \to -1} f(x) = \infty$

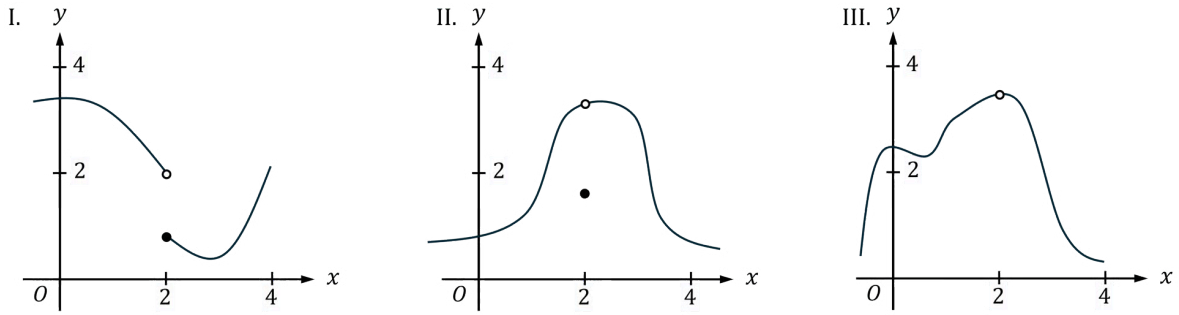
B. $\lim_{x \to 0} f(x) = -1$

C. $\lim_{x \to -\infty} f(x) = -1$

D. $f(x) \neq -1$ for all $x \leq 0$

(1 point)

9 For which of the following does $\lim_{x \to 2} f(x)$ exist?



A. I only

B. II only

C. I and III only

D. II and III only

(1 point)

Hard Questions

1 If $\lim_{x \to 2} f(x) = 5$, which of the following must be true?

I. f is continuous at $x = 2$.

II. f is differentiable at $x = 2$.

III. $f(2) = 5$

A. None

B. I only

C. I and III only

D. I, II and III

(1 point)

2 Let f, g and h be continuous functions such that $\lim_{x \to 5} f(x) = -2$, $\lim_{x \to 3} g(x) = 5$ and $\lim_{x \to 3} h(x) = -7$.

If the function p is defined by $p(x) = \dfrac{f(g(x))}{[h(x)]^2}$, then $\lim_{x \to 3} p(x)$ is

A. $-\dfrac{2}{49}$

B. $\dfrac{5}{49}$

C. $\dfrac{2}{7}$

D. unable to be determined from the information provided

(1 point)

3 $\lim_{x \to 2} \left(\dfrac{\sqrt{x+1} - \sqrt{3}}{x-2} \right) =$

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A. $\frac{0}{0}$

B. $\frac{\sqrt{3}}{6}$

C. $\sqrt{6}$

D. $2\sqrt{3}$

(1 point)

4 Let f and g be the functions defined by $f(x) = x^2 - 2x + 2$ and $g(x) = \sin\left(\frac{\pi x}{2}\right)$. Let h be a function defined for all real numbers and satisfying $f(x) \le h(x) \le g(x)$ for all values of x except $x = 1$.

What is $\lim_{x \to 1} h(x)$?

A. -1

B. 0

C. 1

D. unable to be determined from the information provided

(1 point)

5 If

$$f(x) = \begin{cases} \frac{\sin(7-7x)}{\sin(2x-2)} & \text{for } -\frac{1}{2} \le x < 1 \\ 0 & \text{for } x = 1 \\ \frac{x^2 - 3x + 9}{2} & \text{for } x > 1 \end{cases}$$

then $\lim_{x \to 1} f(x)$ is

A. $-\frac{7}{2}$

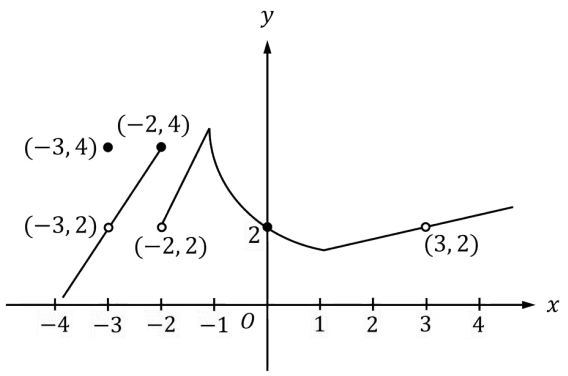
B. 0

C. $\frac{7}{2}$

D. nonexistent

(1 point)

6



Graph of f

The graph of a function f is shown above. For which of the following values of c does $\lim_{x \to c} f(x) = 2$?

A. 0 only

B. -3 and 0 only

C. -3, 0, and 3 only

D. -3, -2, 0 and 3

(1 point)

7 What are all the horizontal asymptotes of the graph of $y = \frac{3 + 5^x}{2 + 5^x}$ in the xy-plane.

A. $y = 1$ only

B. $y = \frac{3}{2}$ only

C. $y = 1$ and $y = \frac{3}{2}$ only

D. $y = 0$ and $y = \frac{3}{2}$ only

(1 point)